

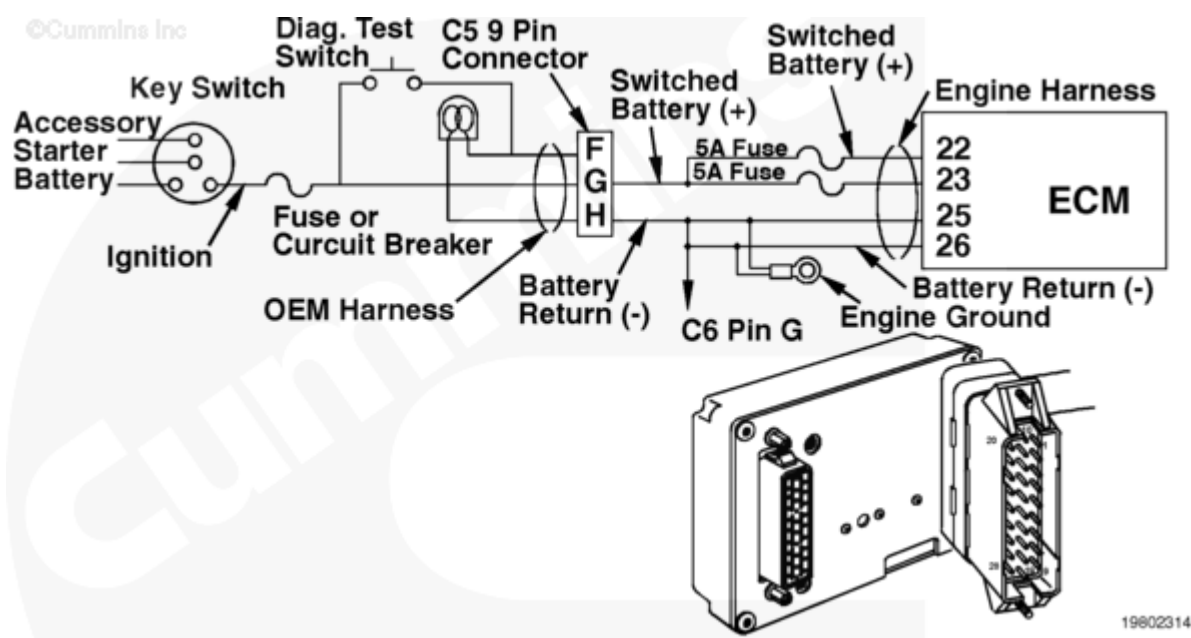
Fault Code: 442

Battery 1 Voltage - Data Valid But Above Normal Operating Range - Moderately Severe Level

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 442 PID(P): S168 SPN: FMI: 3 Lamp: On SRT: 00-636	ECM supply voltage is below the minimum system voltage level.	Engine will run at idle only .



Switched Battery Supply Circuit

Circuit Description

The engine control module (ECM) receives constant voltage from the batteries through the unswitched battery wires that are connected directly to the positive (+) battery post. The ECM receives switched battery input through the keyswitch wire when the keyswitch is turned ON.

Component Location

The location of the battery will vary with the original equipment manufacturer (OEM).

Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected high battery supply voltage.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE lamp immediately when the diagnostic runs and fails.
- Engine will **only** idle.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.
- Engine will **only** idle.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning alternator that is overcharging.
- Batteries connected in series instead of parallel.
- Incorrect jump-starting procedure.
- A weak or damaged battery.

Refer to Troubleshooting Fault Code t05-442 (</qs3/pubsys2/xml/en/procedures/98/98-t05-442.html>)

Last Modified: 10-Sep-2021
